



SCENT WORK 1 E-BOOK

INTRO TO SCENT WORK: Building the love of the hunt.

Week 1

The nosework we are doing in this class is the type used to teach detection dogs. Detection dogs learn to identify one or more specific scents and do not follow a trail or identify disturbed vegetation.

This initial class is an introduction to patterned hunting for your dog. The objective is to introduce it to your dog and to present scent pictures that have an increasing level of difficulty. They will begin by searching for food only. The next stage of training is the pairing of food with another odor and transferring the object of the hunt. But first, dogs must learn that hunting for food is a great game that they always look forward to.

Training format: Boxes are the context cue that the hunt is on. We will begin with 4 to 6 boxes and easy finds the first week and will progress to increasingly difficult scent problems in the succeeding weeks. We will include height, outdoor and vehicle search problems throughout the six weeks. We will work from finding a small bag filled with food to finding a single piece of food.

Homework: For your dog to develop the drive to hunt for the scent, you need to set up practice sessions several times a week at home. If your dog appears to be losing interest, make the scent problems easier until his enthusiasm returns. Try to use the same box for the food find as much as possible to avoid contaminating all of them.

All the materials you will need are a food and a few cardboard boxes. Self-rewarding is the most important aspect of these initial sessions. The goal is to get your dog DRIVING to the food. We want the dogs to want to move mountains to get that treat. Experiment with foods and toys to see what your dog drives to best.

Information about your dog's nose from Alabama A&M and Auburn Universities:

The Dog's Sense of Smell

Olfaction, the act or process of smelling, is a dog's primary special sense. A dog's sense of smell is said to be a thousand times more sensitive than that of humans. In fact, a dog has more than 220 million olfactory receptors in its nose, while humans have only 5 million. Because of this keen sense of smell, dogs are able to locate everything from forensic cadaver material to disaster survivors as demonstrated during the terrorist attack on the World Trade Center on September 11, 2001.

Anatomy

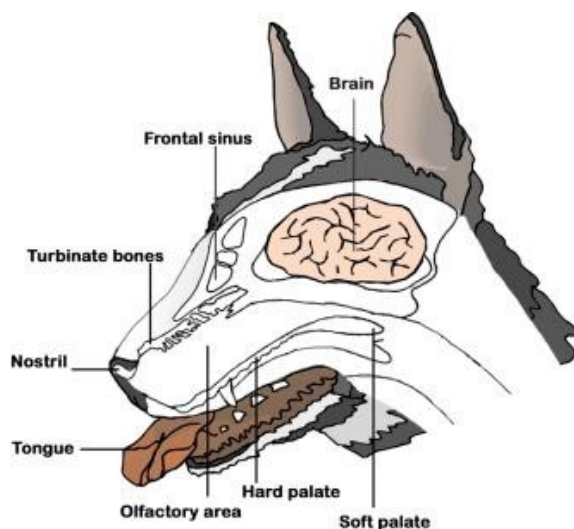
A dog's nose consists of a pair of nostrils (nares) for inhaling air and odors and a nasal cavity. The olfactory receptor cells in a dog's nose extend throughout the entire layer of specialized olfactory epithelium found on the ethmo-turbinate bones of the nasal cavity. The olfactory portion of the nasal mucous membrane contains a rich supply of olfactory nerves that ultimately connect with the highly developed olfactory lobe in the dog's brain.

Interestingly, the olfactory receptors in the nasal cavity are anatomically distinct from those in the vomeronasal organ. Each receptor neuron (nerve cell) in the olfactory epithelium of the nasal cavity has a dendrite that ends in a knob with several thin cilia covered by mucus. Receptor neurons in the vomeronasal organ typically lack cilia but have microvilli on the cell surface. Dogs possess an additional olfactory chamber called the vomeronasal organ that also contains olfactory epithelium. The vomeronasal organ, known as Jacobson's organ, consists of a pair of elongated, fluid-filled sacs that open into either the mouth or the nose. It is located above the roof of the mouth and behind the upper incisors.

Physiology

A dog's nose is normally cool and moist. The moisture secreted by mucous glands in the nasal cavity captures and dissolves molecules in the air and brings them into contact with the specialized olfactory epithelium inside the nose. Dogs use sniffing to maximize detection of odors. The sniff is actually a disruption of the normal breathing pattern. Sniffing is accomplished through a series of rapid, short inhalations and exhalations. A bony subethmoidal shelf, which is found below the ethmoturbinate bones of the nasal cavity, forces inhaled air into the olfactory epithelium. Washing out of the region upon exhalation does not occur due to the nasal pocket created by the bony subethmoidal shelf. The nasal pocket permits the odor molecules that are unrecognizable in a single sniff to accumulate and interact with olfactory receptors. Odor molecules in the olfactory epithelium of the nasal cavity are absorbed into the mucous layer and diffuse to the cilia of receptor neurons. This

interaction generates nerve impulses that are transmitted by the olfactory nerves to the dog's brain, which has a well-developed olfactory lobe. This allows the dog to recognize a scent and follow a trail. Olfactory receptor cells in the send impulses to the region associated with sexual and is believed to be important in pheromones (body scents). for the dog's ability to identify and people.



vomerolateral organ also of the hypothalamus social behaviors. This organ the detection of This theory could account and recognize other animals

Utility

Today, people use a dog's keen sense of smell in many ways. Federal, state, and local government agencies employ specially trained dogs in search and rescue missions and in the detection of narcotics and contraband agriculture products. The Federal Emergency Management Agency has national dog-handler teams that respond to disasters worldwide. State and local law enforcement agencies in the United States (U.S.) have canine units trained to detect drugs and search for lost individuals, homicide victims, and forensic cadaver materials. U.S. Customs and Border Protection has more than 800 canine teams that work with the U.S. Department of Homeland Security to combat terrorist threats, stop the flow of illegal narcotics, and detect unreported currency, concealed humans, or smuggled agriculture products. Its Canine Enforcement Program (CEP) uses a variety of dogs including Labrador retrievers, golden retrievers, German shepherds, Belgian Malinois, and many mixed breeds. The CEP uses beagles to detect agriculture contraband. The passively trained Beagle Brigade dogs detect prohibited fruits, plants, and meats in baggage and vehicles of international travelers as they go through Federal Inspection Service areas. Beagle Brigade teams work at several major border-crossing stations in the United States as well as many international airports that are ports of entry into this country. Medical tests have recently shown that specially trained dogs are capable of detecting certain types of tumors in humans.

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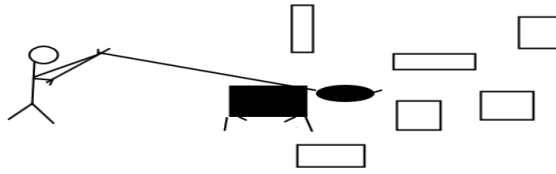
End of Week 1

Week 2

Class exercises:

- 3 to 6 boxes Spread out boxes place them in different shapes to make hunt more difficult. Add other objects, such as buckets or baskets.
- Threshold exercises to create excitement and make easy finds.
- Use rabbit in a hole concept. Make the rabbit tricky and get the box to move if your dog shows hesitation.
- Close some of the box flaps and don't show the dog the food going in (blind hides for the dog, but not for you).

Long line handling: Keep one hand under the line so that the line stays over your dog's back and the other hand close to your body, usually about waist high. It is your stabilizing hand.



Homework: Set up practice sessions several times a week at home. The goal is to get the dog to search enthusiastically without stopping so try to avoid excess talking to your dog. Be sure you are letting the dog do the search-no pointing nor urging. If your dog is having a hard time or looks discouraged, quietly move around the space to get your dog moving and searching for the scent cone.

All hides are still in the boxes. If your dog appears to be losing interest, make the scent problems easier until their enthusiasm returns. This includes using only one container for several repetitions.

Start putting find objects in other types of containers and make food less accessible. Move to other locations. Remember your dog should get success each time by self-rewarding followed by your reward.

Learn to read the dog's change in behavior when they get near the source. This will become your alerting response.

Toys: It is ok to have your dog find a toy. It should be the same toy each time but remember not to name it. A second toy in your hand will allow you to reward quickly. We can pair either food or toys with the target odors.

End of Week 2

Week 3

Class exercises:

Spread out boxes and other containers to make hunt more difficult.

Use threshold (scent right by the entry) exercises to create excitement and make easy finds. Try a hides in a bathroom.

Spreading out the hides to new locations shows more motivation to solve the problems. Keep it exciting. Try hiding in different textures and heights (up to about 2 feet).

Dogs can work on a long line or off leash in smaller safe areas, such as indoors. Avoid trying to control your dog

Take scent/food vessels out with you and do some NBN (Nothing but Nosework) walks.

NBN (nothing but nosework) walks: Take along food and toys. Hold your dog back by the collar or harness while tossing a piece of food or a toy into high grass or another hiding location. Walk your dog in a circle or away from the area then enthusiastically send your dog back to find.

You can do this many time during the walk. It adds fun and stimulation to a regular walk and dogs just love it.

Homework: Set up practice sessions several times a week at home. If your dog appears to lose interest, make the scent problems easier until their enthusiasm returns.

Do some very easy and some more challenging in each session. End on an easy one.

Vary types of containers and make food a bit more inaccessible. Try to work in other locations. Remember your dog should get success each time with both a self reward and a reward from you.

Observe carefully so you learn to read the dog's change in behavior as they get near the source (in Odor). This will start an alerting response. Then wait and watch for their final response when they have pinpointed the find.

Pinpointing scent: Be sure your dog is on source (i.e. pinpointing the scent) before you reward. If you have set up self-rewards this is not an issue but you must get there quickly to reward on source regardless. The source is the highest concentration of odor in the space.

End of Week 3

Week 4

Goals: Determine if your dog is really hunting. Check yourself to be sure you are not over-helping.

Class exercises:

Develop a Plan B-what do I do if my dog can't find it?

Start to use the *alert* call & get to the dogs rapidly to reward on source.

Reduce the number of boxes you use and hide in other types of containers and locations

Set up situations that allow your dog to solve the problem of SOURCING. He/she must find and point out the exact location of the source.

Outdoor Game The bunny is on the run: As soon as your dog finds the hide, grab your container and attach or drop it somewhere else – rapidly giving your dog the go find cue. Try to fool your dog.

Reward just above or at the side of the container.

Homework: Try to get someone to set up a few blind hides. Be careful to give your helper instructions about what your dog can handle so the hunt won't be too frustrating. One blind hide to every 10 known ones is the general rule. You need to know where the hides are.

The new hides should still be set up so the dog can self-reward before you

get there. Work on hanging back to observe your dog's behavior – try to

stay out of your dog's

Try hides in 2 new locations every training session, both exterior and interior, containers and on objects.

You should know where the hides are while you are learning your dog's first alert behavior. Add some elevation but not higher than 2 feet. *Pinpointing scent:* Effective scent work requires that you reward *on source* and that you wait until your dog is on source. Remember that they are learning to locate the highest concentration of odor.

HANDLER ATTITUDE:

Handling a scent work dog requires the handler to be positive and trust their dog. If you have negative attitude about your dog or your abilities as a handler, it will travel down the leash to your dog. Negative thoughts occur when we lack experience or have had a bad experience.

No matter what level you achieve or what your goals are, you and your dog should be having fun. Confront each odor problem with patience and sharp observation. Accept the failure of an exercise as an equal to a success – you get a lot of useful information from it. Attack scent work with the joy and bravado your dog shows. Keep in mind that this is the dog's game!

DEFINITIONS:

Residual odor: The odor that remains when the source is removed

Over saturation: A large, overwhelming source of odor, a level with which your dog is not familiar.

Odor obedience: A canine's overriding and strong desire to get to the source of the odor.

Scent transfer: When odor molecules from a source are transferred by human or animal contact, wind or incidental contact with an object.

Scent cone: A pattern of odor molecules that forms when emanating from a source.

Scent discrimination: The ability to focus on or detect a specific odors when other odors are present.

Threshold of detection: The level of odor that the canine can identify. Professional detection dogs are trained to respond to a variety of levels. *Pinpointing scent:* Be sure you are letting your dog pinpoint the highest concentration of the scent reward religiously *on source*.

End of Scentwork 1

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